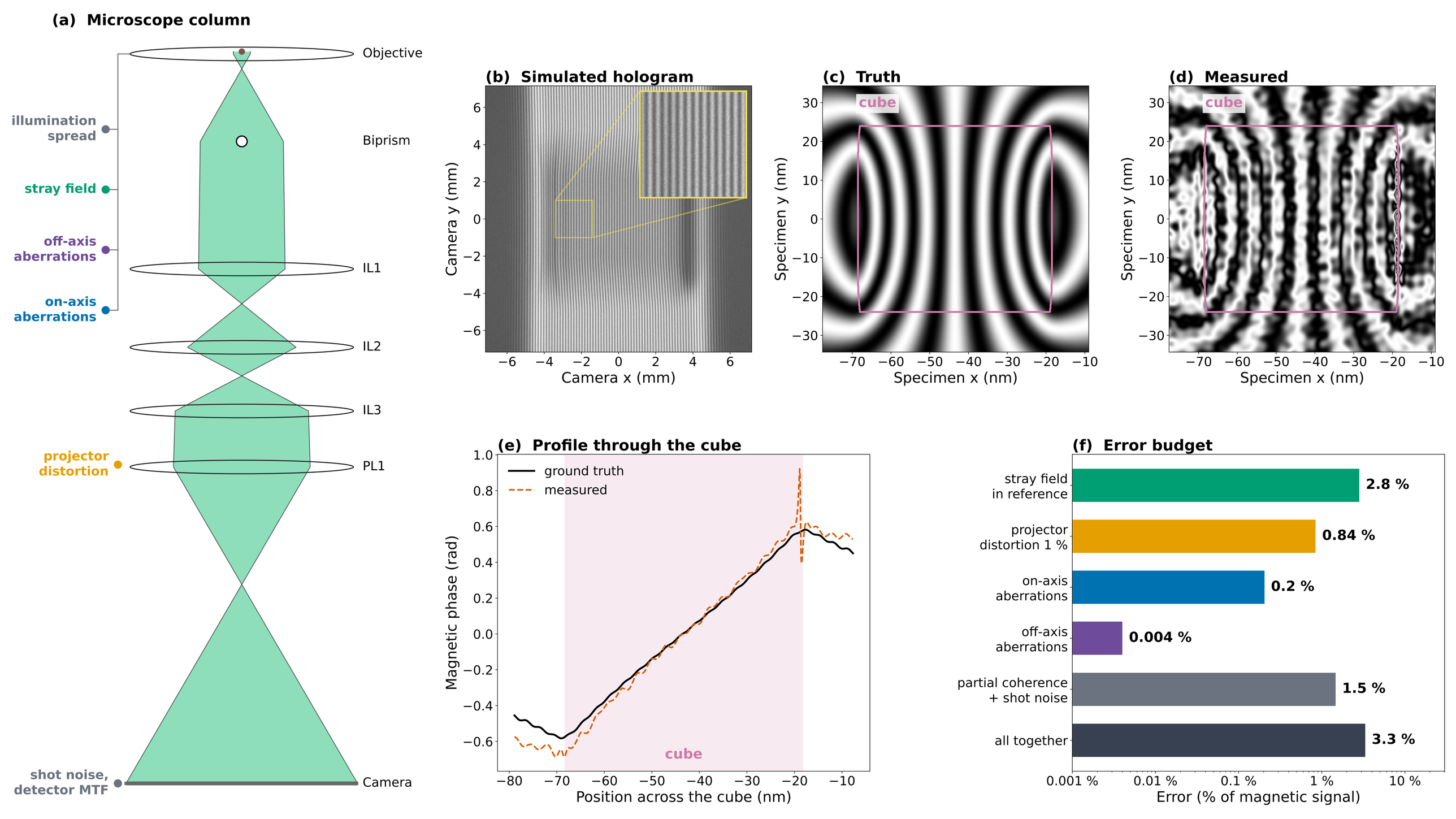

(a) Microscope column
Objective
illumination spread
Biprism
stray field
off-axis aberrations
IL1
on-axis aberrations
IL2
IL3
projector distortion
PL1
shot noise, detector MTF
Camera
(b) Simulated hologram
Camera y (mm)
Camera x (mm)
(c) Truth
cube
Specimen y (nm)
Specimen x (nm)
(d) Measured
cube
Specimen y (nm)
Specimen x (nm)
(e) Profile through the cube
ground truth
measured
cube
Magnetic phase (rad)
Position across the cube (nm)
(f) Error budget
stray field in reference
2.8 %
projector distortion 1 %
0.84 %
on-axis aberrations
0.2 %
off-axis aberrations
0.004 %
partial coherence + shot noise
1.5 %
all together
3.3 %
0.001 %
0.01 %
0.1 %
1 %
10 %
Error (% of magnetic signal)